ModelIt!

Teacher's Guide

Why Can't Weather Apps Get It Right?

Modeling Atmospheric Systems, Chaos Theory, and the Limits of Prediction

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-ESS2-5, HS-ESS2-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-5, HS-ESS2-3

Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes. Develop a model based on evidence of Earth's interior to describe the cycling of matter by thermal convection.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan investigations to test how varying initial conditions in their atmospheric model affect forecast outcomes, demonstrating sensitive dependence.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — The foundation for Earth's global climate system is the electromagnetic radiation from the sun and its interaction with Earth's surface and atmosphere. Weather results from complex interactions among atmospheric variables driven by solar energy.
Crosscutting Concept	**Cause and Effect** — Students investigate how small causes (tiny measurement errors) can produce large, unpredictable effects in complex systems, distinguishing deterministic chaos from randomness.

CAST Testing Alignment

MULTIPLE CHOICE:

A meteorologist runs two computer forecasts with initial temperatures differing by only 0.01°C. The 3-day forecasts are nearly identical, but the 10-day forecasts predict completely different weather. This is best explained by:

CONSTRUCTED RESPONSE:

Using your model, explain why it is mathematically impossible to produce an accurate 30-day weather forecast even with perfect technology. Include the concepts of sensitive dependence, measurement uncertainty, and chaotic systems in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

In 1961, meteorologist Edward Lorenz accidentally discovered one of the most profound principles in science. While running a weather simulation, he rounded an initial value from 0.506127 to 0.506 — a difference of less than 0.02%. The resulting forecast was completely different. This discovery — sensitive dependence on initial conditions — revealed that the atmosphere is a chaotic system with a fundamental prediction limit that no technology can ever overcome.

How Weather Works

All weather is driven by differential solar heating. The equator receives more solar energy per square meter than the poles, creating temperature gradients that drive atmospheric circulation. Warm air rises (convection), creating low pressure at the surface. Cool air sinks, creating high pressure. Air flows from high to low pressure (wind), carrying moisture evaporated from oceans. When moist air rises and cools, water vapor condenses into clouds and precipitation. This cycle — heating, convection, moisture transport, condensation — produces all weather.

Chaos Theory and Weather

Chaotic systems are deterministic (governed by fixed physical laws) but unpredictable beyond a certain time horizon because they exhibit sensitive dependence on initial conditions. In the atmosphere, this means that two nearly identical starting states will diverge exponentially over time. The 'Lyapunov time' for the atmosphere — how long before small errors double — is approximately 2-3 days. After 5 doublings (~10-15 days), initial measurement errors have grown to the size of the weather features themselves, making the forecast meaningless.

Modern Weather Forecasting

Today's forecasts use numerical weather prediction (NWP): supercomputers solve the Navier-Stokes equations and thermodynamic equations on a 3D grid covering the globe. Models divide the atmosphere into millions of cells, each about 3-10 km across and hundreds of meters tall. Every cell's temperature, pressure, humidity, and wind are updated every few minutes. To address chaos, forecasters run 'ensemble models' — 20-50 slightly different versions of the same forecast — and analyze how much they diverge to estimate confidence.

Forecast Accuracy by Lead Time

Modern 1-day forecasts are about 95% accurate — essentially as good as they can get. 3-day forecasts are about 90% accurate, comparable to what 1-day forecasts were in the 1980s. 5-day forecasts are about 80% accurate, rivaling 3-day forecasts from 20 years ago. Beyond 10 days, accuracy drops below 60%. Beyond 14 days, forecasts provide only marginal improvement over using historical climate averages. The theoretical maximum useful forecast for the atmosphere is approximately 2-3 weeks.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify solar radiation input, air temperature, moisture content, wind speed, and storm probability as the key components of the atmospheric weather system.

E **Establish Relationships**
Students determine that solar radiation drives temperature, temperature differences drive wind, wind transports moisture, and the interaction of all four factors determines storm probability.

V **Visualize & Model**
Students build a computational model showing how atmospheric variables interact to produce weather patterns and how tiny changes in initial conditions cascade into different forecasts.

E **Evaluate Outcomes**
Students run stable weather, approaching front, and butterfly effect scenarios to discover the limits of predictability in chaotic atmospheric systems.

R **Revise & Extend**
Students add pressure gradient, wind shear, or land-ocean contrast to model more realistic atmospheric dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your weather app says 70% chance of rain on Saturday. Saturday comes — it's sunny. Why? The answer involves one of the deepest discoveries in mathematics."*

Watch For

Quick poll: How far into the future do you trust a weather forecast? 1 day? 3 days? 7 days? Record responses.

Slide 2: Learning Objectives

3 min

Say: *"Today we're discovering WHY weather is so hard to predict — and it's not because meteorologists are bad at their jobs."*

Watch For

Have students read objectives. Pre-teach 'atmospheric convection' and 'sensitive dependence' with visual demonstrations.

Slide 3: Big Question

3 min

Say: *"We have satellites watching every inch of the planet and supercomputers doing quadrillions of calculations per second. So why can't we get a 14-day forecast right?"*

Watch For

Students hypothesize: Is it bad technology? Not enough data? Or something more fundamental?

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of the atmosphere and then deliberately breaking it to discover the limits of predictability."*

Watch For

Preview LEVER steps. Emphasize that we're testing the LIMITS of the model, not just how it works.

Slide 5: Activity 1: Components

8 min

Say: *"What factors drive weather? Which one is the ultimate energy source for ALL atmospheric motion?"*

Watch For

Guide sorting. Solar radiation is the only external input — everything else responds to it. Discuss why.

Slide 6: Activity 2: Connections

8 min

Say: *"The sun heats the surface unevenly. That creates temperature differences. Temperature differences create wind. Wind carries moisture. Moisture creates storms. It's all connected."*

Watch For

Build the energy cascade from solar input to storms. Emphasize that every variable affects every other.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now the fun part — let's change ONE number by 0.1% and see what happens to the 10-day forecast."*

Watch For

Students predict whether 0.1% matters. Run twin simulations. Watch the forecasts diverge. This is the butterfly effect.

Slide 8: Discoveries

5 min

Say: *"Edward Lorenz discovered this in 1961 by accident. He rounded a number from 0.506127 to 0.506, and the entire weather forecast changed. That 0.02% difference rewrote the future."*

Watch For

Show the accuracy decay curve: 95% at 1 day, 80% at 5 days, 50% at 10 days. Discuss the theoretical 2-3 week limit.

Slide 9: STEM Challenge

12 min

Say: *"If you can't predict far ahead, you need to detect FAST. Design a system that gives your school maximum warning time for severe storms."*

Watch For

Groups design alert systems specifying sensors, thresholds, communication channels, and false alarm mitigation.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Solar Radiation Input
- **INTERNAL:** Air Temperature, Moisture Content, Wind Speed, Storm Probability

Relationships

- **Solar Radiation Input to Air Temperature:** POSITIVE (+) — More solar radiation heats the surface and the air in contact with it. Differential heating by latitude, land/ocean, and cloud cover creates the temperature gradients that drive all atmospheric motion.
- **Air Temperature to Wind Speed:** POSITIVE (+) — Greater temperature differences between air masses create steeper pressure gradients, which drive stronger winds. Temperature is the engine; wind is the motion. Without temperature differences, there would be no wind.
- **Moisture Content to Storm Probability:** POSITIVE (+) — Higher moisture content provides more raw material for precipitation and more latent heat release during condensation. When moist air rises and cools, water vapor condenses, releasing energy that fuels thunderstorm updrafts, creating a positive feedback loop that intensifies storms.

Key Discoveries

- Weather is driven by energy transfers — solar radiation heats the surface unevenly, creating temperature differences that drive wind, which carries moisture, which produces precipitation
- The atmosphere is a chaotic system — tiny measurement errors in initial conditions amplify over time, making long-range forecasts fundamentally unreliable beyond about 10-14 days
- Storm formation requires the simultaneous interaction of multiple factors: instability, moisture, lifting mechanism, and wind shear — missing any one ingredient dramatically reduces storm probability
- Weather prediction accuracy decreases predictably: 1-day forecasts are about 95% accurate, 5-day about 80%, 10-day about 50%, and beyond 14 days, forecasts provide little useful information

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Weather apps are inaccurate because meteorologists are bad at their jobs"	Modern weather forecasting is one of the greatest achievements of computational science. A 5-day forecast today is as accurate as a 1-day forecast was in the 1980s. The remaining inaccuracy isn't due to incompetence — it's due to chaos theory, a fundamental mathematical property of the atmospheric equations. No amount of skill can overcome the butterfly effect beyond approximately 2 weeks.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If we had enough data and computing power, we could predict weather perfectly"	This is the fundamental insight of chaos theory: perfect prediction requires perfect measurements of initial conditions, which is physically impossible. The atmosphere has approximately 10^44 molecules, each with position and velocity that would need to be measured with infinite precision. Even quantum mechanics (Heisenberg's uncertainty principle) prevents measurements below certain precision limits. Better computers and data improve short-range forecasts but cannot extend the fundamental prediction horizon.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Climate can't be predicted if weather can't"	Weather and climate are fundamentally different prediction problems. Weather is about the specific state of the atmosphere on a specific day — chaotic and unpredictable beyond ~2 weeks. Climate is about the statistical average of weather over decades — driven by well-understood energy balance physics and highly predictable. You can't predict which coin flip will be heads, but you can confidently predict that 1,000 flips will be close to 50% heads.